

KALIMATE

< Calcium polystyrene sulfonate >

Storage	Store below 30 °C, in tight container.
Shelf life	5-years

CONTRAINDICATIONS (KALIMATE is contraindicated in the following patients.)

- 1) Patients with intestinal obstruction [Intestinal perforation may occur.]
- 2) Hypersensitivity to polystyrene sulphonate.
- 3) Should not be used in patients with renal failure together with hypercalcaemia.

DESCRIPTION

Composition	Calcium polystyrene sulfonate
Dosage form	Powder
Color	Pale yellowish white to light yellow
Odor	Odorless
Taste	Tasteless

INDICATIONS

Hyperkalemia resulting from acute or chronic renal failure.

DOSAGE AND ADMINISTRATION

KALIMATE is for adult use only.

1. Oral administration

For adults, the daily dosage is 15 - 30 g to be taken dividing into 2 - 3 times. Each dose should be suspended in 30 - 50 mL of water and administered orally. The dosage may be adjusted according to the patient's condition.

Administer calcium polystyrene sulfonate at least 3 hours before or 3 hours after other oral medications. Patients with gastroparesis may require a 6-hour separation (see section DRUG INTERACTIONS).

2. Rectal route

A single dose of 30 g should be suspended in 100 mL of water or a 2% methylcellulose solution and administered via the rectal route after warming to body temperature. It should be left in the intestinal tract for 30 minutes to 1 hour after administration. In case the suspension leaks out, the hip should be lifted up by placing a pillow underneath or the patient should be sit on the knee-chest position. Following retention of the enema, the colon should be irrigated to remove the resin. 5% glucose solution may be substituted for the water or 2% methylcellulose solution.

SYMPTOMS AND TREATMENT FOR OVER DOSAGE, AND ANTIDOTES

Decrease of serum potassium level and increase of serum calcium level may be caused by overdose. If these abnormal findings in electrolytes are observed, appropriate measures such as reducing the dose or withdrawing of the drug should be taken.

PRECAUTIONS¹⁻⁸⁾**1. Careful Administration (KALIMATE should be administered with care in the following patients.)**

- (1) Patients susceptible to constipation [Intestinal obstruction or intestinal perforation may occur.]
- (2) Patients with intestinal stenosis [Intestinal obstruction or intestinal perforation may occur.]
- (3) Patients with gastrointestinal ulcers [Symptoms may be exacerbated.]
- (4) Patients with hyperparathyroidism [Blood concentration of calcium may be increased by ion exchange.]
- (5) Patients with multiple myeloma [Blood concentration of calcium may be increased by ion exchange.]

2. Important Precautions

- (1) Intestinal perforation, intestinal obstruction or large intestinal ulcer may occur. If any abnormal findings such as **severe constipation, prolonged abdominal pain, vomiting or melaena etc.** are observed, administration of the drug should be discontinued and appropriate measures taken.
- (2) Patients should be instructed to pay attention to their feces and consult with their physicians if constipation is followed by significant symptoms such as abdominal pain, abdominal distention, or vomiting, etc.
- (3) This product should be administered while measuring the serum potassium and serum calcium levels regularly to prevent the overdose. If any abnormal findings are observed, appropriate measures such as reducing the dose or withdrawing of the drug should be taken.

3. Drug Interactions

Precautions for coadministration (KALIMATE should be administered with care when coadministered with the following drugs.)

Drugs	Signs, Symptoms, and Treatment	Mechanism and Risk Factors
Digitalis preparation Digoxin etc.	Digitalis intoxication effect may be enhanced.	The enhancement is due to the serum potassium lowering effect of this product.
Antacid and laxative containing aluminum, magnesium or calcium Dried aluminum hydroxide gel, Magnesium hydroxide, Precipitated calcium carbonate, etc.	The effect of this product may be reduced. It has been reported that coadministration with the drugs mentioned left caused systemic alkalosis etc. ²⁻⁴⁾	There is a possibility that this product exchanges randomly for cations in the drugs mentioned left. The neutralization of hydrogencarbonate secreted in the intestinal tract is prevented. ²⁾
Thyroid hormone drugs Levothyroxine, etc.	The effects of thyroid hormone drugs may be reduced. It has been reported this inhibition effect may be diminished by administering thyroid hormone drugs and this drug at different times.	This drug may adsorb thyroid hormone drugs in the intestinal tract, thus inhibit the intake of these drugs into the body.

Calcium polystyrene sulfonate, as with other cation-exchange resin i.e. sodium polystyrene sulfonate, has the potential to bind other drugs. In vitro binding studies, sodium polystyrene sulfonate was shown to significantly bind the oral medications (warfarin, metoprolol, phenytoin, furosemide, amlodipine and amoxicillin) that were tested. Binding of sodium polystyrene sulfonate to other oral medications could cause decreased gastrointestinal absorption and loss of efficacy when taken close to the time sodium polystyrene sulfonate is administered. As with sodium polystyrene sulfonate, administer calcium polystyrene sulfonate at least 3 hours before or 3 hours after other oral medications. Patients with gastroparesis may require a 6-hour separation (see section DOSAGE AND ADMINISTRATION).

4. Adverse Reactions

169 cases of adverse reactions to this drug were reported in 151 (12.8%) of 1,182 patients with oral administration in clinical studies before and after drug approval. The major adverse reactions were constipation in 109 patients (9.2%), anorexia in 18 patients (1.5%), nausea in 16 patients (1.4%), etc. Changes in laboratory data were observed in 13 patients (1.1%). These changes were hypopotassemia, which can be normalized by adjustment of the dosage. <At the end of the investigation of adverse reaction incidence>⁵⁾

(1) Clinically significant adverse reactions

Since intestinal perforation, intestinal obstruction or large intestinal ulcer may occur (incidence unknown), the patient should be carefully observed. If any abnormalities suggestive of these conditions such as clinically significant constipation, prolonged abdominal pain, vomiting or melaena etc. are observed, administration of the drug should be discontinued and appropriate measures including auscultation, palpation, and diagnostic imaging taken.¹⁾

(2) Other adverse reactions

	≥5%	5% > ≥0.1%	Incidence unknown
Hypersensitivity			Rash
Gastrointestinal	Constipation (Oral administration) ^{Note)}	Nausea, anorexia, stomach discomfort (Oral administration)	Constipation (Rectal route), Diarrhea
Electrolytic		Hypopotassemia (Oral administration)	Hypopotassemia (Rectal route)

Note) See "Important Precautions" (1)

5. Use in the Elderly

Since elderly patients often have reduced physiological function, careful supervision and measures such as reducing the dose are recommended.

6. Use during Pregnancy, Delivery or Lactation

The safety of this product in pregnant woman has not been established.

7. Precautions concerning Use

(1) Oral administration:

- 1) It has been reported that the oral administration of sorbitol suspension of a similar drug (sodium polystyrene sulfonate) may cause perforation in intestine, necrosis in mucous membrane of intestine, large intestine ulcer, colon necrosis, etc.
- 2) To avoid accumulation of the drug in the gastrointestinal tract following oral administration, care should be taken to prevent constipation.

(2) Rectal route:

- 1) Animal studies (rat) have shown that the administration of sorbitol via rectal route may cause intestinal necrosis.^{6,7)} It has also been reported outside Japan that the administration via rectal route of sorbitol suspension of a cation exchange resin of polystyrene sulfonate type may cause colon necrosis. In case of the administration via rectal route, sorbitol solution should not be used.⁶⁻⁸⁾
- 2) To avoid retention of the drug in the intestinal tract following rectal administration, ensure that the drug is excreted. Particularly in patients experiencing difficulty excreting naturally, appropriate measures should be taken to remove the drug from the intestinal tract.

8. Other Precautions

- (1) It has been reported that cases of colon stenosis and colon ulcer, etc. occurred after oral administration of sorbitol suspension of KALIMATE.
- (2) Forming of insoluble gel-like substance in the intestinal tract has been reported when coadministered this drug with sodium alginate.

PHARMACOKINETICS^{9, 10)}

(For reference)

It is thought that this product is not absorbed (rabbit).⁹⁾ However, it has been reported from the result of the experiment with calf that fine particles less than 5 µm in size has been absorbed through mucous membrane and deposited on the reticuloendothelial system tissue.¹⁰⁾ KALIMATE is controlled in order that such fine particles are less than 0.1% in rate.

PHARMACOLOGY¹¹⁻¹⁵⁾

1. Pharmacological action

- (1) KALIMATE contains 7.0 - 9.0% calcium, 1 g of which is exchanged for 53 - 71 mg (1.36 - 1.82 mEq/g) of potassium in vitro (KCl solution).
- (2) By administering 15 - 30 g/day of KALIMATE to renal failure patients (adults), the serum potassium level is reduced about 1 mEq/L (humans).¹¹⁻¹³⁾
- (3) Unlike sodium-typed resin, KALIMATE does not cause the increase in serum sodium and phosphate levels and the decrease in serum calcium level when it is used for the renal failure (humans).¹¹⁻¹³⁾
- (4) Since KALIMATE is calcium-typed resin, it is used even for the patients who restricted ingestion of sodium. Moreover, it can be used without fear of appearance and aggravation of edema, hypertension or cardiac insufficiency induced by sodium (humans).^{14, 15)}

2. Mechanism of action

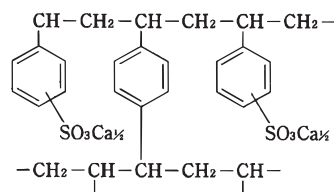
After administration of KALIMATE via oral or rectal route, calcium ion of KALIMATE is exchanged for potassium ion in the intestinal tract, particularly around the colon, and the drug is excreted as unchanged polystyrene sulfonate resin into the feces without digestion and absorption. In consequence, potassium in the intestinal tracts excreted outside the body.

PHYSICOCHEMISTRY

Nonproprietary name: Calcium polystyrene sulfonate (JAN)

Structural formula:

It has an irregularly intricate sterical structure. A part of it is shown as follows.



Description:

It occurs as a pale yellowish white to light yellow powder with odorless and tasteless. It is practically insoluble in water, ethanol and ether.

PACKAGING

Powder: 5 g × 126 sachets/box, 500 g/bottle

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